Pilot Analysis

==============================

LOAD DATA

==============================

```
d_rds = readRDS(here("data","processed","clean_pilot_data_zr.rds"))
d = d_rds[keep_flag == 1]
```

==============================

DESCRIPTIVE STATISTICS

==============================

```
vars <- c("expert_trust_index", "health_engagement_index",
"meat_exp_cred", "meat_infl_cred", "oil_exp_cred", "oil_infl_cred",
"oil_cred", "meat_cred")

stargazer( as.data.frame(d[, ..vars]), type = "latex", header = FALSE,
title = "Descriptive Statistics for Pilot Variables", summary.stat =
c("mean","sd","min","max") )
```

Table 1: Descriptive Statistics for Pilot Variables

Statistic	Mean	St. Dev.	Min	Max
expert_trust_index	5.013	1.044	2.600	7.000
health_engagement_index	5.258	1.374	1.800	7.000
meat_exp_cred	4.750	1.584	1.000	7.000
meat_infl_cred	5.404	1.327	3.333	7.000
oil_exp_cred	5.000	1.232	3.667	7.000
oil_infl_cred	4.528	1.720	1.000	7.000
oil_cred	4.817	1.432	1.000	7.000
meat_cred	5.151	1.442	1.000	7.000

```r
lapply(d[, ..vars], function(x) sum(is.na(x)))
```

$expert_trust_index [1] 0

$health_engagement_index [1] 0

$meat_exp_cred [1] 19

$meat_infl_cred [1] 12

$oil_exp_cred [1] 12

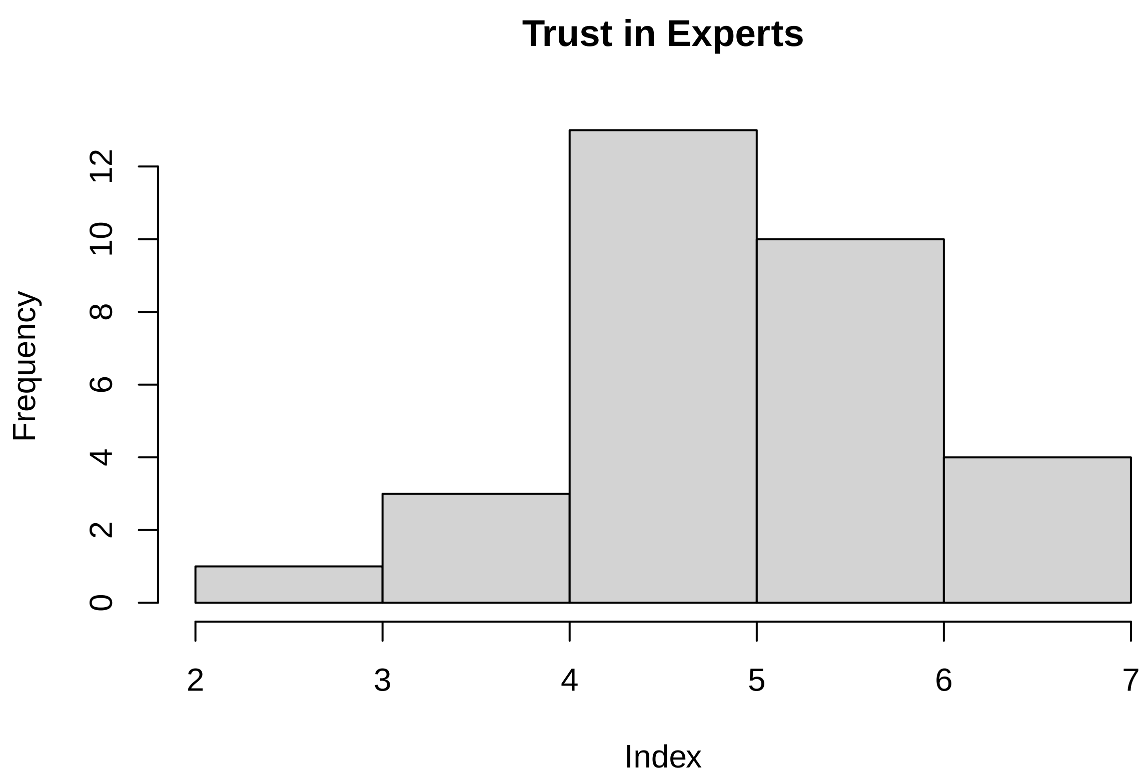
$oil_infl_cred [1] 19

$oil_cred [1] 0

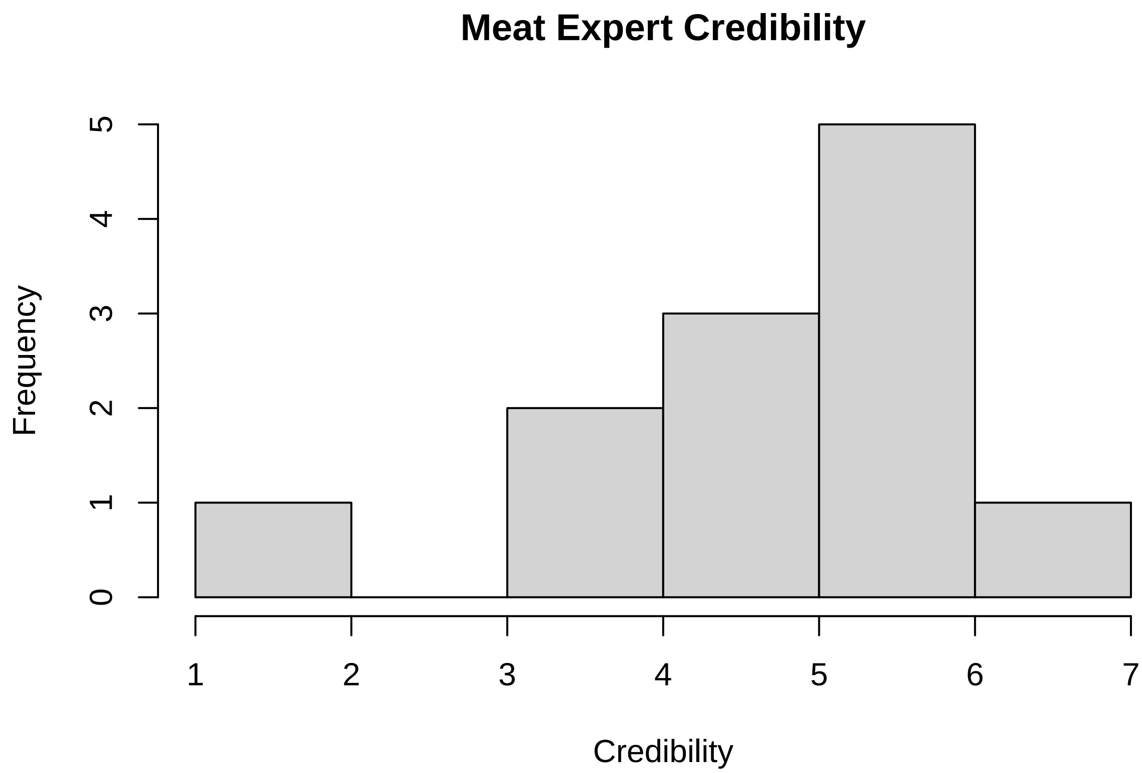
$meat_cred [1] 0

```r
# ========================
# HISTOGRAMS
# ========================
```

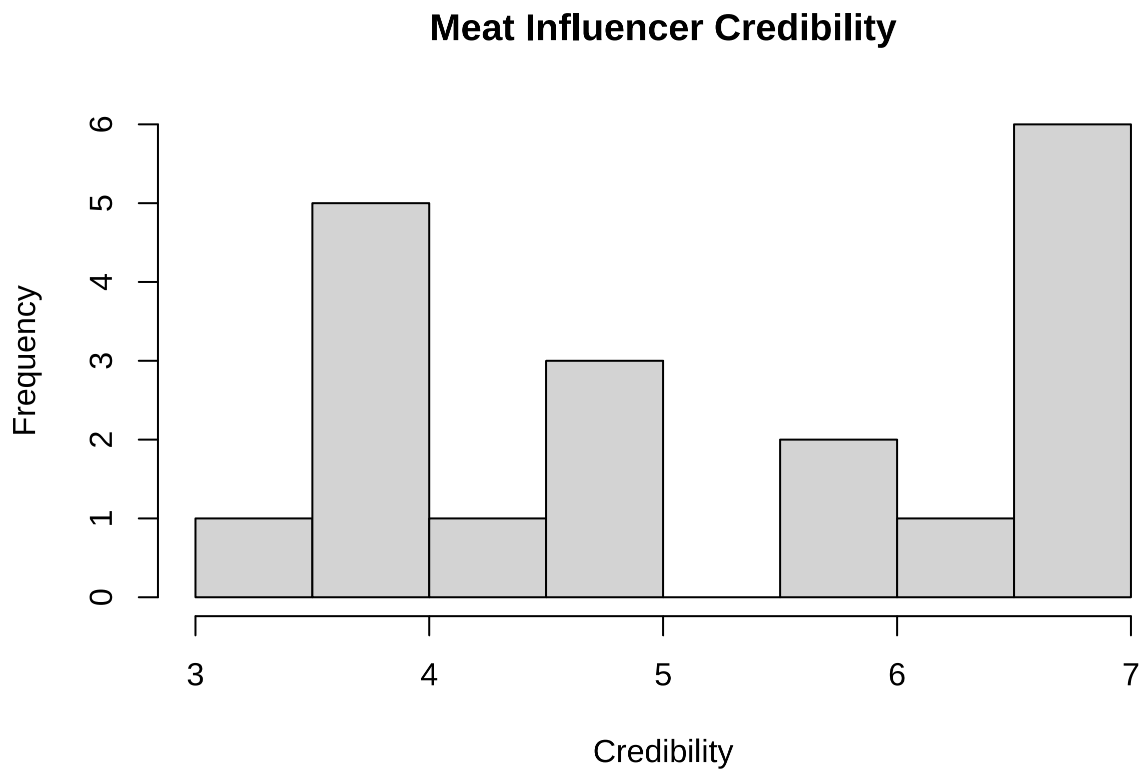
```r
hist(d$expert_trust_index, main = "Trust in Experts", xlab = "Index")
```



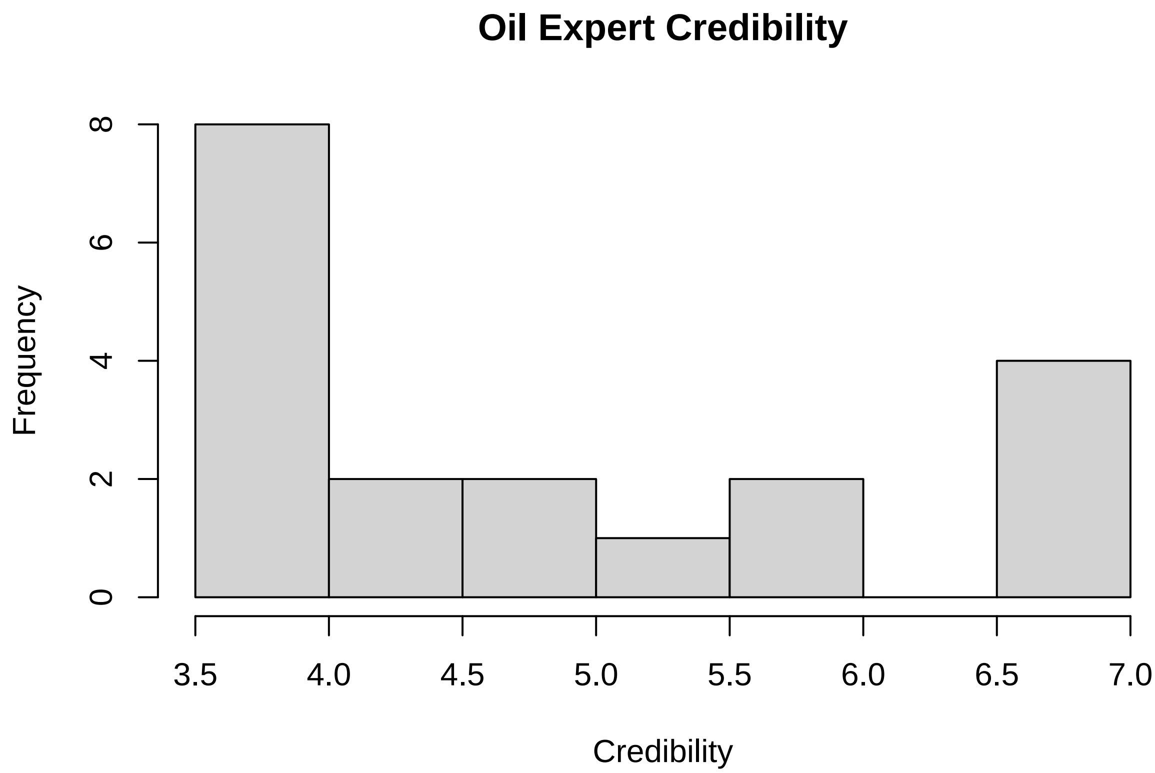

```r
hist(d$meat_exp_cred, main = "Meat Expert Credibility", xlab = "Credibility")
```



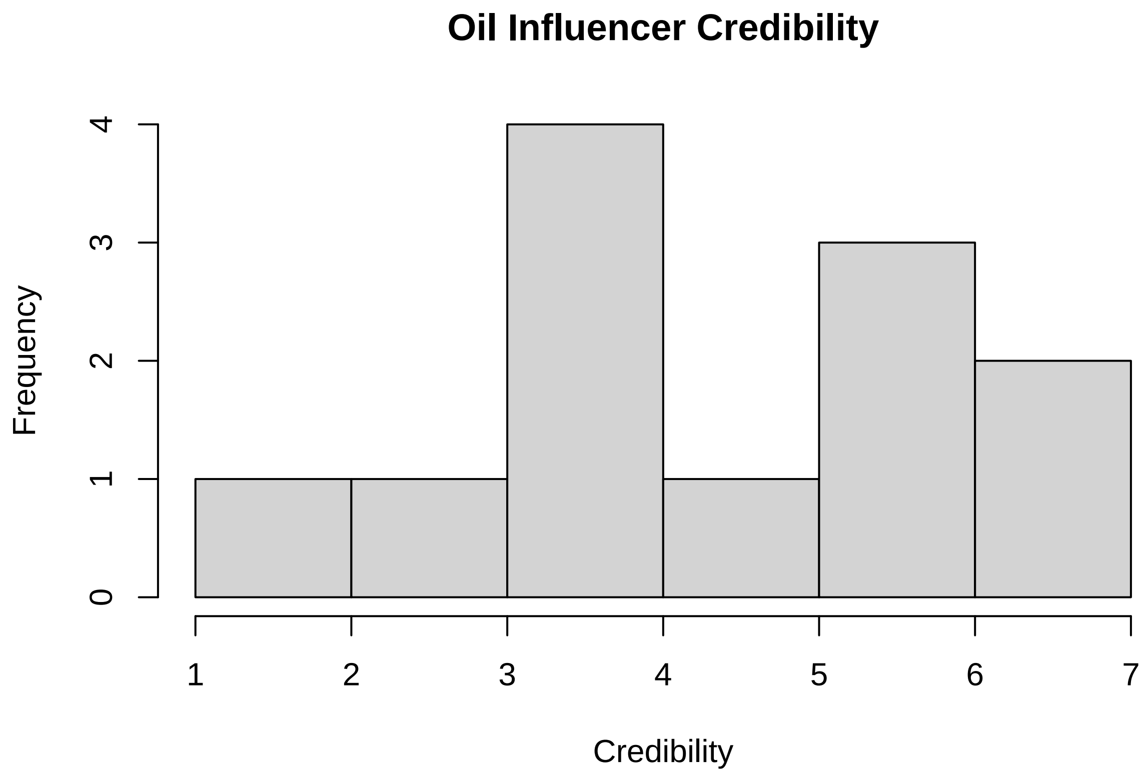

```r
hist(d$meat_infl_cred, main = "Meat Influencer Credibility", xlab = "Credibility")
```




```r
hist(d$oil_exp_cred, main = "Oil Expert Credibility", xlab = "Credibility")
```




```r
hist(d$oil_infl_cred, main = "Oil Influencer Credibility", xlab = "Credibility")
```



==============================

PAIRED T-TESTS

==============================

```
t_meat <- t.test(d$meat_exp_cred, d$meat_infl_cred, paired =
FALSE)
print(t_meat)
```

```
##
##  Welch Two Sample t-test
##
## data:  d$meat_exp_cred and d$meat_infl_cred
## t = -1.19, df = 20.458, p-value = 0.2477
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -1.7974538  0.4904362
## sample estimates:
## mean of x mean of y
##  4.750000  5.403509
```

```r
t_oil <- t.test(d$oil_exp_cred, d$oil_infl_cred, paired = FALSE)
print(t_oil)
```

```
##
##  Welch Two Sample t-test
##
## data:  d$oil_exp_cred and d$oil_infl_cred
## t = 0.82645, df = 18.123, p-value = 0.4193
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.7276356  1.6720800
## sample estimates:
## mean of x mean of y
##  5.000000  4.527778
```

```
# ========================
# REGRESSION ANALYSIS
# ========================
```

```r
    model_oil_1 <- lm(oil_cred ~ post_oil_source, data = d)
    coeftest(model_oil_1, vcov = vcovHC(model_oil_1, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)                5.00000    0.28449 17.5750   <2e-16 ***
## post_oil_sourceinfluencer -0.47222    0.56793 -0.8315   0.4125
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
    model_oil_2 <- lm(oil_cred ~ post_oil_source + expert_trust_index + health_engagement_index, data =
    coeftest(model_oil_2, vcov = vcovHC(model_oil_2, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)                0.766966   1.022206  0.7503  0.45956
## post_oil_sourceinfluencer -0.094672   0.446589 -0.2120  0.83371
## expert_trust_index         0.562993   0.243930  2.3080  0.02889 *
## health_engagement_index    0.240518   0.184193  1.3058  0.20264
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
    model_oil_3 <- lm(oil_cred ~ post_oil_source + expert_trust_index + health_engagement_index + age +
    coeftest(model_oil_3, vcov = vcovHC(model_oil_3, type = "HC1"))
```

```
##
## t test of coefficients:
```

```
##
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)              1.608665   2.356615  0.6826  0.50268
## post_oil_sourceinfluencer -0.161416   0.501776 -0.3217  0.75103
## expert_trust_index       0.475851   0.270667  1.7581  0.09403 .
## health_engagement_index    0.215020   0.181634  1.1838  0.25037
## age18-24                 0.300014   0.934954  0.3209  0.75163
## age25-34                 1.141438   0.771405  1.4797  0.15454
## age35-44                -0.658138   1.020626 -0.6448  0.52636
## age45-54                -0.704754   0.831175 -0.8479  0.40653
## age55-64                -0.062616   0.798259 -0.0784  0.93826
## gender                   0.146304   0.581545  0.2516  0.80393
## education               -0.135526   0.224147 -0.6046  0.55222
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
model_oil_4 <- lm(oil_cred ~ post_oil_source * expert_trust_index + health_engagement_index, data =
coeftest(model_oil_4, vcov = vcovHC(model_oil_4, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                                           Estimate Std. Error t value
## (Intercept)                                1.16686    1.10266  1.0582
## post_oil_sourceinfluencer                 -1.28428    2.81910 -0.4556
## expert_trust_index                         0.51217    0.26274  1.9493
## health_engagement_index                    0.21559    0.18055  1.1941
## post_oil_sourceinfluencer:expert_trust_index  0.24111    0.56005  0.4305
##                                           Pr(>|t|)
## (Intercept)                                0.29969
## post_oil_sourceinfluencer                  0.65248
## expert_trust_index                         0.06212 .
## health_engagement_index                    0.24323
## post_oil_sourceinfluencer:expert_trust_index  0.67037
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
model_oil_5 <- lm(oil_cred ~ post_oil_source * (expert_trust_index + health_engagement_index), data
coeftest(model_oil_5, vcov = vcovHC(model_oil_5, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                                                  Estimate Std. Error t value
## (Intercept)                                     1.3404397  1.1738537  1.1419
## post_oil_sourceinfluencer                      -1.3501579  2.8518002 -0.4734
## expert_trust_index                              0.6142729  0.2826711  2.1731
## health_engagement_index                         0.0888848  0.2471812  0.3596
## post_oil_sourceinfluencer:expert_trust_index   -0.0018336  0.6571033 -0.0028
## post_oil_sourceinfluencer:health_engagement_index  0.2438038  0.3783008  0.6445
##                                                 Pr(>|t|)
## (Intercept)                                     0.26431
## post_oil_sourceinfluencer                       0.64001
```

```
## expert_trust_index                                   0.03945 *
## health_engagement_index                              0.72217
## post_oil_sourceinfluencer:expert_trust_index         0.99780
## post_oil_sourceinfluencer:health_engagement_index    0.52514
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
model_meat_1 <- lm(meat_cred ~ post_meat_source, data = d)
coeftest(model_meat_1, vcov = vcovHC(model_meat_1, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                            Estimate Std. Error t value  Pr(>|t|)
## (Intercept)                 4.75000    0.45251  10.497 2.171e-11 ***
## post_meat_sourceinfluencer  0.65351    0.54643   1.196    0.2414
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
model_meat_2 <- lm(meat_cred ~ post_meat_source + expert_trust_index + health_engagement_index, data
coeftest(model_meat_2, vcov = vcovHC(model_meat_2, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 1.22948    1.23016  0.9994 0.326454
## post_meat_sourceinfluencer  0.21269    0.44505  0.4779 0.636573
## expert_trust_index          0.20541    0.22055  0.9314 0.359918
## health_engagement_index     0.52510    0.17990  2.9188 0.007004 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
model_meat_3 <- lm(meat_cred ~ post_meat_source + expert_trust_index + health_engagement_index + age
coeftest(model_meat_3, vcov = vcovHC(model_meat_3, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                             Estimate Std. Error t value Pr(>|t|)
## (Intercept)                -0.281350   2.181592 -0.1290 0.898673
## post_meat_sourceinfluencer  0.224464   0.396794  0.5657 0.577895
## expert_trust_index          0.177169   0.261905  0.6765 0.506494
## health_engagement_index     0.522823   0.163068  3.2062 0.004433 **
## age18-24                    1.634281   0.893182  1.8297 0.082238 .
## age25-34                    2.090513   0.927104  2.2549 0.035506 *
## age35-44                    1.224770   1.063802  1.1513 0.263182
## age45-54                    1.253627   0.988317  1.2684 0.219205
## age55-64                    1.323725   0.971838  1.3621 0.188316
## gender                      0.107450   0.506014  0.2123 0.833986
## education                   0.041807   0.198884  0.2102 0.835633
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
model_meat_4 <- lm(meat_cred ~ post_meat_source * expert_trust_index + health_engagement_index, data
coeftest(model_meat_4, vcov = vcovHC(model_meat_4, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                                          Estimate Std. Error t value
## (Intercept)                               2.94443    2.36248  1.2463
## post_meat_sourceinfluencer               -2.37068    2.42522 -0.9775
## expert_trust_index                       -0.20781    0.47644 -0.4362
## health_engagement_index                   0.57923    0.18187  3.1849
## post_meat_sourceinfluencer:expert_trust_index  0.52359    0.46456  1.1271
##                                          Pr(>|t|)
## (Intercept)                               0.22375
## post_meat_sourceinfluencer                0.33732
## expert_trust_index                        0.66631
## health_engagement_index                   0.00374 **
## post_meat_sourceinfluencer:expert_trust_index  0.27001
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
model_meat_5 <- lm(meat_cred ~ post_meat_source * (expert_trust_index + health_engagement_index), da
coeftest(model_meat_5, vcov = vcovHC(model_meat_5, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                                                 Estimate Std. Error t value
## (Intercept)                                     2.953023   2.396155  1.2324
## post_meat_sourceinfluencer                     -2.365424   2.473351 -0.9564
## expert_trust_index                             -0.219044   0.543716 -0.4029
## health_engagement_index                         0.588566   0.309583  1.9012
## post_meat_sourceinfluencer:expert_trust_index       0.542970   0.581762  0.9333
## post_meat_sourceinfluencer:health_engagement_index -0.019447   0.366249 -0.0531
##                                                Pr(>|t|)
## (Intercept)                                     0.22926
## post_meat_sourceinfluencer                      0.34805
## expert_trust_index                              0.69047
## health_engagement_index                         0.06887 .
## post_meat_sourceinfluencer:expert_trust_index       0.35958
## post_meat_sourceinfluencer:health_engagement_index  0.95808
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
stargazer(model_meat_1, model_meat_2, model_meat_3, model_meat_4,
          type = "latex")
```

% Table created by stargazer v.5.2.3 by Marek Hlavac, Social Policy Institute. E-mail: marek.hlavac at gmail.com % Date and time: Tue, Apr 07, 2026 - 20:02:04

Table 2:

	Dependent variable:		
	meat_cred		
	(1)	(2)	(3)
post_meat_sourceinfluencer	0.654	0.213	0.224
	(0.527)	(0.445)	(0.506)
expert_trust_index		0.205	0.177
		(0.298)	(0.338)
health_engagement_index		0.525^{**}	0.523^{*}
		(0.230)	(0.273)
age18-24			1.634
			(1.128)
age25-34			2.091^{*}
			(1.162)
age35-44			1.225
			(1.340)
age45-54			1.254
			(1.088)
age55-64			1.324
			(1.116)
gender			0.107
			(0.548)
education			0.042
			(0.266)
post_meat_sourceinfluencer:expert_trust_index			
Constant	4.750^{***}	1.229	-0.281
	(0.413)	(1.054)	(2.355)
Observations	31	31	31
R^2	0.050	0.408	0.507
Adjusted R^2	0.018	0.342	0.260
Residual Std. Error	1.430 (df = 29)	1.170 (df = 27)	1.241 (df = 20)
F Statistic	1.537 (df = 1; 29)	6.204^{***} (df = 3; 27)	2.053^{*} (df = 10; 20)
Note:			*p<0.1;

```r
stargazer(model_oil_1, model_oil_2, model_oil_3, model_oil_4, model_oil_5,
          type = "latex")
```

% Table created by stargazer v.5.2.3 by Marek Hlavac, Social Policy Institute. E-mail: marek.hlavac at gmail.com % Date and time: Tue, Apr 07, 2026 - 20:02:04

==============================

ATTRITION ANALYSIS

==============================

```r
summary(d_rds$complete)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.0000  0.0000  0.0000  0.4842  1.0000  1.0000
```

```r
table(d_rds$complete)
```

```
##
##  0  1
## 49 46
```

```r
mean(d_rds$complete == 1, na.rm = TRUE)
```

```
## [1] 0.4842105
```

```r
table(d_rds$post_meat_source, d_rds$complete)
```

```
##
##               0  1
##   expert      3 24
##   influencer  3 22
```

```r
prop.table(table(d_rds$post_meat_source, d_rds$complete), 1)
```

```
##
##                      0         1
##   expert     0.1111111 0.8888889
##   influencer 0.1200000 0.8800000
```

```r
t.test(complete ~ post_meat_source, data = d_rds)
```

Table 3:

	Dependent variable:		
	oil_cred		
	(1)	(2)	(3)
post_oil_sourceinfluencer	−0.472	−0.095	−0.161
	(0.530)	(0.457)	(0.493)
expert_trust_index		0.563*	0.476
		(0.305)	(0.329)
health_engagement_index		0.241	0.215
		(0.236)	(0.265)
age18-24			0.300
			(1.098)
age25-34			1.141
			(1.130)
age35-44			−0.658
			(1.304)
age45-54			−0.705
			(1.058)
age55-64			−0.063
			(1.086)
gender			0.146
			(0.533)
education			−0.136
			(0.258)
post_oil_sourceinfluencer:expert_trust_index			
post_oil_sourceinfluencer:health_engagement_index			
Constant	5.000***	0.767	1.609
	(0.330)	(1.146)	(2.302)
Observations	31	31	31
R^2	0.027	0.369	0.526
Adjusted R^2	−0.007	0.299	0.289
Residual Std. Error	1.437 (df = 29)	1.199 (df = 27)	1.207 (df = 20)
F Statistic	0.794 (df = 1; 29)	5.268*** (df = 3; 27)	2.222* (df = 10; 20)

Note:

```
##
##  Welch Two Sample t-test
##
## data:  complete by post_meat_source
## t = 0.098169, df = 49.365, p-value = 0.9222
## alternative hypothesis: true difference in means between group expert and group influencer is not equ
## 95 percent confidence interval:
##  -0.1730371  0.1908149
## sample estimates:
##     mean in group expert mean in group influencer
##                0.8888889                0.8800000
```

```
lm(complete ~post_meat_source, data = d_rds)
```

```
##
## Call:
## lm(formula = complete ~ post_meat_source, data = d_rds)
##
## Coefficients:
##                (Intercept)  post_meat_sourceinfluencer
##                   0.888889                   -0.008889
```

```
table(d_rds$post_oil_source, d_rds$complete)
```

```
##
##               0  1
##   expert      3 22
##   influencer  3 24
```

```
prop.table(table(d_rds$post_oil_source, d_rds$complete), 1)
```

```
##
##                      0         1
##   expert     0.1200000 0.8800000
##   influencer 0.1111111 0.8888889
```

```
t.test(complete ~ post_oil_source, data = d_rds)
```

```
##
##  Welch Two Sample t-test
##
## data:  complete by post_oil_source
## t = -0.098169, df = 49.365, p-value = 0.9222
## alternative hypothesis: true difference in means between group expert and group influencer is not equ
## 95 percent confidence interval:
##  -0.1908149  0.1730371
## sample estimates:
##     mean in group expert mean in group influencer
##                0.8800000                0.8888889
```

```r
lm(complete ~post_oil_source, data = d_rds)
```

```
##
## Call:
## lm(formula = complete ~ post_oil_source, data = d_rds)
##
## Coefficients:
##               (Intercept)  post_oil_sourceinfluencer
##                  0.880000                   0.008889
```

```r
lm(complete ~ post_meat_source + age + gender + expert_trust_index,
data = d_rds)
```

```
##
## Call:
## lm(formula = complete ~ post_meat_source + age + gender + expert_trust_index,
##     data = d_rds)
##
## Coefficients:
##                (Intercept)  post_meat_sourceinfluencer
##                    0.88770                    -0.02069
##                   age18-24                    age25-34
##                   -0.13210                    -0.23311
##                   age35-44                    age45-54
##                   -0.21991                    -0.12262
##                   age55-64                      gender
##                   -0.01487                    -0.06945
##         expert_trust_index
##                    0.05038
```

```r
lm(complete ~ post_oil_source + age + gender +
expert_trust_index, data = d_rds)
```

```
##
## Call:
## lm(formula = complete ~ post_oil_source + age + gender + expert_trust_index,
##     data = d_rds)
##
## Coefficients:
##               (Intercept)  post_oil_sourceinfluencer
##                   0.86701                    0.02069
##                  age18-24                   age25-34
##                  -0.13210                   -0.23311
##                  age35-44                   age45-54
##                  -0.21991                   -0.12262
##                  age55-64                     gender
##                  -0.01487                   -0.06945
##        expert_trust_index
##                   0.05038
```

```r
table(d_rds$keep_flag)
```

```
##
##  0  1
## 64 31
```

```r
table(d_rds$keep_flag, d_rds$complete)
```

```
##
##      0  1
##   0 49 15
##   1  0 31
```

```r
prop.table(table(d_rds$keep_flag, d_rds$complete), 1)
```

```
##
##            0        1
##   0 0.765625 0.234375
##   1 0.000000 1.000000
```